Detailed Methodology — Urban Agriculture and Urban Food Security in Nigeria

From frozen API downloads to spatial regression: equations, layman explanations, and why each step matters

Jerry Zachary Bannister — Statistician, Abubakar Tafawa Balewa University (ATBU), Bauchi, Nigeria

16 October 2026

Table of contents

1 Overview and reproducibility contract 2
1.1 Pipeline at a glance 2
2 Data acquisition: what, how, why 3
2.1 2.1 World Bank Indicators API — national backdrop 3
2.2 2.2 OpenStreetMap Overpass API — urban agriculture footprints 4
2.3 2.3 NASA POWER API — agro-climate 4
2.4 2.4 FAOSTAT API — food-security stub 5
2.5 2.5 GADM administrative boundaries — spatial framework 5
3 Cleaning and preparation (`analysis/spatial/01_clean.R`) 5
3.1 3.1 OSM → `sf` points → GeoPackage 5
3.2 3.2 World Bank long → wide 5
3.3 3.3 NASA daily → annual 6
4 Spatial framework and descriptive 6
4.1 4.1 Area and density 6
4.2 4.2 Join and count 6
5 Spatial dependence: Moran's I and LISA 7
5.1 5.1 Weight matrix $\mathbf{W}$ 7
5.2 5.2 Global Moran's I 7
5.3 5.3 LISA 7
6 Spatial regression: from OLS to lag/error 8
6.1 6.1 Why OLS is biased 8
6.2 6.2 Lag and error specifications 8
6.3 6.3 Estimates and model selection 8
6.3.a 6.3.1 Superseded: Gaussian spatial models on a zero-inflated ratio 8
6.3.b 6.3.2 The corrected model: counts with an area exposure 9
6.3.c 6.3.3 LISA with multiple-comparison control 10
6.3.d 6.4 The climate hypothesis (H2) cannot be tested with these data 10
6.3.e 6.5 The OSM-completeness confound (open) 11
7 Additional analyses 11
7.1 7.1 Density vs climate (descriptive only) and scale 11
7.2 7.2 Maps 12
8 Reproducibility, ethics, reflection 12
9 How to re-run 12

10 References 13
Bibliography 13

How to use this document

This is the **technical companion** to the manuscript, `manuscript/manuscript.qmd`, which is now formatted for the *African Journal on Land Policy and Geospatial Sciences* (AJLP&GS, ISSN 2657-2664) rather than for the earlier conference. It documents **every step from frozen API downloads to final maps and regressions**, with **formal equations** and **plain-English boxes** explaining *what* was done and *why*. All numbers trace to `data/raw/provenance.json` and `scripts/fetch_*.py` → `analysis/spatial/*.R` on `R 4.5.2` / `ds-general` Python 3.12. The manuscript is the concise narrative; this file is the reproducible audit trail.

1 Overview and reproducibility contract

1.1 Pipeline at a glance



In plain English: why freeze?

APIs change daily. If we analysed live data, results would not be reproducible. We **query each API once**, save the response to `data/raw/*.csv`, and record the URL, timestamp and licence in `data/raw/provenance.json`. All later analysis reads **only the frozen CSVs**. Anyone can re-run the analysis without network access and get the same numbers.

Environments (see `ENVIRONMENTS.md`): `R 4.5.2` with `sf 1.1.2` (GEOS 3.14.1/GDAL 3.12.1/PROJ 9.7.1, `sf_use_s2(TRUE)`), `spdep 1.4.2`, `spatialreg 1.4.3`, `terra 1.9.46`, `dplyr 1.2.1`, `ggplot2 4.0.3`; Python `ds-general` 3.12 (`pandas` 2.2.3, `requests` 2.34.2, `numpy` 1.26.4). Random seeds set (`set.seed(1)`). Git repo `D:\urban-agriculture-nigeria-spatial` ↔ GitHub `batestguy/urban-agriculture-nigeria-spatial` (branch `master`).

Study Area: Five cities maximising agro-ecological variation — Abuja (planned capital, Guinean savanna), Lagos (coastal megacity), Kano (Sahelian), Port Harcourt (humid Niger Delta), Gombe (savanna, host). LGAs (GADM v4.1, 775 units) are the areal unit. Hypotheses: **H1** density is clustered ($I > 0$); **H2** density increases with aridity (lower PRECTOT, higher T2M) — see the caveat in §6.4, this hypothesis turns out to be **not testable** with the frozen climate data; **H3** density declines with LGA area.

2 Data acquisition: what, how, why

2.1 2.1 World Bank Indicators API — national backdrop

What: Three indicators for Nigeria (`NG`) 2000–2024:

- `SP.URB.TOTL` — urban population (total)
- `AG.PRD.FOOD.XD` — food production index (2004–06 = 100)
- `SN.ITK.DEFC.ZS` — prevalence of undernourishment (%)

How: `scripts/fetch_worldbank.py` loops `https://api.worldbank.org/v2/country/NG/indicator/{code}?format=json&per_page=100&date=2000:2024`, parses JSON `value`, writes long-format `data/raw/worldbank_ng.csv` (75 rows, 25 per indicator, 2 NAs per indicator) and appends to `provenance.json` with URL, `n`, `timestamp_utc`, `license: CC BY-4.0`.

In plain English

The World Bank API is the safest starting point (no auth, stable). We began here to verify the pipeline before riskier Overpass calls. National aggregates cannot be joined to LGAs, so they are used as *context* (urbanisation 146.5M in 2024, food index 119.9 in 2022, undernourishment 19.9% in 2023) rather than outcomes; LGA-level undernourishment will come from DHS/NLSS in extensions [1]. The indicator values themselves are cited to the World Bank's own API rather than to a printed volume: the pull is recorded in `data/raw/provenance.json` with URL, timestamp, row count and CC BY-4.0 licence, and `data/raw/worldbank_ng.csv` is the frozen copy the analysis reads.

Why these indicators: They directly operationalise *urban food security* (population pressure, production, undernourishment) at the only openly-available temporal resolution for Nigeria.

Failure mode: If `SN.ITK.DEFC.ZS` were >50% NA, we would drop it as outcome — here only 2 NAs, usable.

2.2 2.2 OpenStreetMap Overpass API — urban agriculture footprints

What: `landuse` ∈ {allotments, farmland, farmyard} + `leisure=garden` — OSM's tagging for cultivated urban land. Five city bounding boxes (Abuja, Lagos, Kano, Port Harcourt, Gombe).

How: `scripts/fetch_osm_urban_ag.py` builds **per-tag** queries (not a 4-tag union) to avoid Overpass `504`/`406`:

```
[out:json][timeout:30];
(
  nwr["landuse"="allotments"](bbox);
  nwr["landuse"="farmland"](bbox);
  nwr["landuse"="farmyard"](bbox);
  nwr["leisure"="garden"](bbox);
);
out center tags;
```

Executed sequentially with `sleep 1` per tag, `sleep 2` per city, three fallback endpoints (`overpass-api.de`, `kumi.systems`, `openstreetmap.fr`), `User-Agent: urban-agriculture-nigeria-spatial/1.0`, `Content-Type` header fixed (initial 406 error). Deduplicated by `osm_type+osm_id`, giving **484 rows** (Kano 189, Lagos 130, Port Harcourt 88, Gombe 45, Abuja 32) plus per-city CSVs and `osm_gombe_test.csv` (45 rows test). Provenance ODbL, 5 city entries + 3 zero-row retries (406⟶504 mitigation) in `provenance.json`.

> In plain English: why per-tag?
>
> A single query asking for all four tags at once timed out (90s, `504`) and sometimes returned `406 Not Acceptable` without a proper `User-Agent`. Splitting into four small queries (30s each) and trying three servers reliably returns data. Deduplication ensures a farm tagged both `farmland` and `farmyard` is counted once.

Why OSM: Only openly-licensed, georeferenced source for *intra-urban* cultivated patches at LGA resolution; completeness varies (Kano highest, Abuja lowest) — documented as limitation and mitigated by density normalisation and Kano-excluded sensitivity.

2.3 2.3 NASA POWER API — agro-climate

What: Daily `T2M` (°C), `PRECTOTCORR` (mm, bias-corrected), `RH2M` (%) for 2023 (`20230101–20231231`), `community=AG`, for five city centroids:

- Abuja 9.0765°N, 7.3986°E; Lagos 6.5244, 3.3792; Kano 12.0022, 8.592; Port Harcourt 4.8156, 7.0498; Gombe 10.2896, 11.1677.

How: `scripts/fetch_nasa_power.py` loops `https://power.larc.nasa.gov/api/temporal/daily/point?parameters=T2M,PRECTOTCORR,RH2M&community=AG&latitude={lat}&longitude={lon}&start=20230101&end=20231231&format=JSON`, writes `data/raw/nasa_power_{city}.csv` (365 rows each, 1,825 total). Provenance 5 entries. Annual means/totals in `data/processed/nasa_annual.csv` via `analysis/spatial/01_clean.R`: Abuja 25.9°C/1,031 mm/70.1%, Gombe 27.1/862/53.4, Kano 27.3/404/42.1, Lagos 26.8/1,726/85.4, Port Harcourt 26.2/1,891/87.7.

In plain English

NASA POWER's AG community product is designed for agricultural applications and is validated for West Africa by the POWER project's own release documentation. That documentation is not indexed in Crossref, so it is cited here by its canonical location rather than given a fabricated bibliographic entry: https://power.larc.nasa.gov/docs/methodology/. The 2023 window matches the OSM freeze year; a multi-year window is the next step. Contrasts (404 mm Kano vs 1,891 mm Port Harcourt) provide the gradient to test **H2** (aridity).

2.4 2.4 FAOSTAT API — food-security stub

What: `https://fenixservices.fao.org/faostat/api/v1/en/data/FS?area=159` (Nigeria).

How: `scripts/fetch_fao.py` → HTTP 521 during freeze → retained **1-row stub** `data/raw/fao_food_security.csv` with `status: http_521_stub` in `provenance.json` and note to replace with manual `data/external/faostat_food_security.csv` per `data/external/README.md`. Explicitly documented as limitation; not used in models.

2.5 2.5 GADM administrative boundaries — spatial framework

What: GADM v4.1 `https://geodata.ucdavis.edu/gadm/gadm4.1/shp/gadm41_NGA_shp.zip` (3.47 MB zip) → `gadm41_NGA_2.shp` (LGA level 2) + `.dbf` + `.shx` + `.prj`/`.cpg`.

How: Downloaded, extracted to `data/external/gadm41_NGA_2.*` (4.56 MB `.shp` + 0.13 MB `.dbf`, 5.1 MB in R, <50 MB → committed), verified `sf::st_read` → 775 LGAs (nominal 774, one extra documented), 14 columns (`GID_2`, `GID_0`, `COUNTRY`, `GID_1`, `NAME_1`, `NAME_2`…), CRS `EPSG:4326` (`st_crs`), `st_is_valid` 775/775. Provenance 16th entry `CC BY 4.0`. Level 0/1 shapefiles deleted to keep repo lean.

In plain English: why GADM?

GADM is the standard open administrative boundary for Nigeria (774 LGAs, here 775 due to version artefact). Using LGAs makes results *actionable* — zoning decisions are made at LGA level — and matches the resolution of `spatialreg` areal models.

3 Cleaning and preparation (`analysis/spatial/01_clean.R`)

3.1 3.1 OSM → `sf` points → GeoPackage

```
osm <- read_csv("data/raw/osm_urban_ag.csv")
osm_sf <- st_as_sf(filter(osm, !is.na(lat), !is.na(lon)), coords=c("lon","lat"),
crs=4326)
st_write(osm_sf, "data/processed/osm_urban_ag.gpkg")
```

Filters NA coordinates, projects to WGS84, writes 484 points.

3.2 3.2 World Bank long → wide

```
wb_wide <- wb %>% filter(!is.na(value)) %>% select(indicator, date, value) %>%
  pivot_wider(names_from=indicator, values_from=value)
write_csv(wb_wide, "data/processed/worldbank_ng_wide.csv")
```

Gives 25 rows (2000–2024) × 3 indicators.

3.3 3.3 NASA daily → annual

```
nasa_all <- bind_rows(lapply(Sys.glob("data/raw/nasa_power_*.csv"), function(f){
  city <- sub(".*nasa_power_(.*)\\.csv","\\1", basename(f))
  df <- read_csv(f); names(df) <- tolower(names(df)); df$city <- city; df
}))
nasa_annual <- nasa_all %>% group_by(city) %>%
  summarise(n_days=n(), t2m_mean=mean(t2m, na.rm=TRUE),
            prectot_sum=sum(prectotcorr, na.rm=TRUE),
            rh2m_mean=mean(rh2m, na.rm=TRUE))
write_csv(nasa_annual, "data/processed/nasa_annual.csv")
```

Annual means/totals for Table 2. They were also mapped to LGA covariates via state (FCT→Abuja, Kano→Kano, Lagos→Lagos, Rivers→Port Harcourt, Gombe→Gombe) for the 34 LGAs with ≥1 point — **but this mapping is the defect described in §6.4 and the resulting models are no longer reported as evidence.** Rainfall sampled at a city centroid cannot vary within that city, so it cannot be an LGA-level covariate.

> In plain English: why these aggregates?
>
> Daily climate is too noisy for an annual study, so summarising to *mean temperature* and *total rainfall* is reasonable — **at the city level**. The error was then treating that city constant as if it varied across the LGAs of a city. Assigning 34 LGAs the rainfall of their city gives 5 unique predictor values, not 34, and inflates the apparent sample size by a factor of about 7. See §6.4.

4 Spatial framework and descriptive

4.1 4.1 Area and density

$$\text{density}_i = \frac{n_i}{\text{area}_{i,km^2}}, \quad \text{area}_{i,km^2} = \frac{\text{st_area}\ (\text{LGA}_i)}{10^6}$$

Computed with `sf` GEOS spherical geometry (appropriate for 4–13°N). LGA areas 11.6–10,358 km² (median 697 km²); ten-fold variation means raw counts n conflate intensity with size — density corrects this, and $I_{counts} = 0.158$ rises to $I_{density} = 0.300$ upon normalisation.

> In plain English
>
> A large rural LGA could have 10 allotments spread thinly, while a small urban LGA has 10 allotments densely packed. Counts treat them equally; density reveals the urban concentration that matters for zoning. Tarauni (23.6 km², 1.06 km^{-2}) vs Ikorodu (369.5 km², 0.14 km^{-2}) illustrates why density, not counts, is policy-relevant.

4.2 4.2 Join and count

```
joined <- st_join(lga, osm)  # left join: each LGA gets its intersecting points
dens <- joined %>% st_drop_geometry() %>% filter(!is.na(city)) %>%
  count(GID_2) %>% rename(lga=GID_2, n=n)
lga <- left_join(lga, dens, by=c("GID_2"="lga")); lga$n[is.na(lga$n)] <- 0
```

```
lga$area_km2 <- as.numeric(st_area(lga))/1e6
lga$density <- lga$n / pmax(lga$area_km2, 1)
```

Bug audit: Early `count(NAME_2)` spuriously gave 775 LGAs ≥1 point (all); corrected to `count(GID_2)` after `filter(!is.na(city))` ⟶ valid 39 LGAs (5.0%) — documented in `PROGRESS.md`.

5 Spatial dependence: Moran's I and LISA

5.1 5.1 Weight matrix W

$$w_{ij} = \begin{cases} 1 \text{ if LGAs } \ i, j \ \text{ share edge or vertex (queen)} \\ 0 \text{ otherwise} \end{cases}, \qquad w^*_{ij} = w_{ij} / \sum_j w_{ij}$$

Row-standardised queen contiguity via `spdep::poly2nb(queen=TRUE)` ⟶ `nb2listw(style="W")`. Rook (edge only) yields $I_{density} = 0.306$ vs queen 0.300 (both $z > 17.5$, $p < 0.001$), confirming robustness.

> In plain English
>
> **W** encodes *who is neighbour to whom*. Queen counts diagonal neighbours (like a queen in chess); rook counts only edge neighbours. Row-standardising means each LGA's neighbours sum to 1, so a LGA with many neighbours does not dominate.

5.2 5.2 Global Moran's I

$$I = \frac{N}{W} \frac{\sum_i \sum_j w^*_{ij}(x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2}, \quad E[I] = -\frac{1}{N-1}, \quad z = \frac{I - E[I]}{\sqrt{\text{Var}\ [I]}}$$

Tested with `spdep::moran.test` (randomisation, `zero.policy=TRUE`). $N = 775$, x is density or counts. Results: counts $I = 0.158$ ($z = 8.47$, $p < 0.001$), **density $I = 0.300$ ($z = 17.61$, $p < 0.001$)** — strong clustering, rejecting complete spatial randomness (CSR). Excluding Kano State ($N = 731$) retains $I = 0.274$ ($z = 16.3$, $p < 0.001$), so H1 is not Kano-driven.

> In plain English
>
> Moran's I asks: *do similar values cluster?* $I \approx 0$ means random scatter; $I > 0$ means high values neighbour high, low neighbour low. Here $I = 0.300$ is far above the $E[I] = -0.001$ expected under randomness, with $z = 17.61$ — overwhelming evidence of clustering. Density clusters more than counts, proving area normalisation matters.

5.3 5.3 LISA

$$I_i = \frac{x_i - \bar{x}}{m_2} \sum_j w^*_{ij}(x_j - \bar{x}), \quad m_2 = N^{-1} \sum_i (x_i - \bar{x})^2$$

`spdep::localmoran` gives I_i and p_i; quadrants HH/LL/HL/LH classified by $(x_i > \bar{x}, \text{lag}_i > \bar{\text{lag}})$ only when $p_i < 0.05$, mapped in `results/figures/lisa_map.png` (HH red, LL blue). HH centres on Kano metropolis (Ungogo, Tarauni, Kumbotso ring) and Obio/Akpor (Rivers); LL spans sparsely-mapped savanna; HL/LH <10. Stored as `lisa_I`, `lisa_p`, `lisa_quad`, `lag_density` in `results/lga_lisa.gpkg`.

6 Spatial regression: from OLS to lag/error

6.1 6.1 Why OLS is biased

OLS assumes independent residuals. Table 3 rejects CSR, so residuals remain spatially autocorrelated, violating Gauss–Markov and biasing standard errors.

6.2 6.2 Lag and error specifications

$$y = \rho \mathbf{W} y + X\beta + \varepsilon$$

Eq.4 is the **spatial lag**: neighbours' y directly affects y (spillover).

$$y = X\beta + u, u = \lambda \mathbf{W} u + \varepsilon, \varepsilon \sim N(0, \sigma^2 I)$$

Eq.5 is the **spatial error**: omitted spatially-structured covariates induce correlated errors.

Estimated by ML (`spatialreg` 1.4.3). Model choice via AIC, likelihood-ratio $LR = 2(\ell_{spatial} - \ell_{OLS})$, and Lagrange Multiplier LM_{lag} vs LM_{error} and robust RLM; residual Moran reported.

> In plain English
>
> *Lag* says allotment density in a LGA depends on density in neighbouring LGAs (spillover). *Error* says something unmeasured (e.g., soil, market access) is spatially clustered and leaks into residuals. Both correct OLS by explicitly modelling $\mathbf{W}$.

6.3 6.3 Estimates and model selection

> Revised 15 October 2026 — read this before the tables
>
> This section previously reported a `density ~ log(area)` Gaussian fit as the primary density model and a 34-LGA climate model as evidence for **H2**. An audit found two defects in that approach, both documented in `docs/statistical-review-2026-10-15.md`:
>
> 1. **Wrong family.** Density is exactly zero for 95% of the 775 LGAs (skewness 15.5). A Gaussian fit produced **265 of 775 negative fitted densities** — impossible quantities.
> 2. **Pseudoreplication.** The "34 LGA" model used rainfall sampled at **five city centroids**, so the predictor had **5 unique values** (rainfall R^2 on city dummies = 1.000). At the level the data actually replicate — 5 cities — the rainfall effect is $p = 0.58$, not $p = 0.018$.
>
> The Gaussian tables below are retained **for transparency** and are labelled as superseded. The corrected analysis is in §6.3.2 and §6.4, produced by `06_count_models.R` and `08_climate_honest.R`. Every number still reproduces; what changed is which model is believed.

6.3.a 6.3.1 Superseded: Gaussian spatial models on a zero-inflated ratio

Table 5 (superseded). Density $y = n/$ area regressed on log(area), $N = 775$.

Model	log(area)	ρ / λ	AIC
OLS	−0.0086 (0.0014)***	—	−2485.2
Lag (Eq.4)	−0.0052 (0.0013)***	$\rho = 0.419$***	**-2577.5**
Error (Eq.5)	−0.0058 (0.0017)***	$\lambda = 0.419$***	−2572.9

SE in parentheses; ** $p < 0.001$; $LR = 94.25$**. *All three figures reproduce exactly from the frozen data. Two things they do establish and one they do not:*

- **They establish spatial dependence.** Residual Moran falls from **0.256 (OLS) to −0.036 (lag)** — the spatial term removed the dependence OLS left behind. This is a clean result.
- **They establish H3′s direction.** Larger LGAs dilute density, negative and highly significant.
- **They do not establish a trustworthy magnitude**, because regressing a ratio on its own denominator partly recovers a mechanical relationship (the Kronmal ratio fallacy), and **34.2% of fitted values are negative**.

6.3.b 6.3.2 The corrected model: counts with an area exposure

The outcome is a **count** (n_i, features in LGA i) with unequal exposure (area). The right specification puts area in as an offset, not a regressor:

$$n_i \sim \text{NegBin}\ (\mu_i, \theta), \qquad \log \mu_i = \log(\text{area}_i) + X_i\beta$$

Then $\exp(\beta)$ is an **incidence rate ratio** — the density effect, on the correct scale.

Table 5c. Model comparison, $N = 775$.

Model	AIC	Note
Poisson	3569.3	dispersion 39.1 — unusable
NegBin, area as covariate	571.6	counts scale with area
NegBin, pure rate (offset only)	601.7	no area slope
NegBin, rate + area slope (offset)	**571.6**	the defensible model
Hurdle NegBin	534.4	*count component unidentified — see below*
Zero-inflated NegBin	575.6	

Table 5d. The area effect under the correct family.

Term	b	SE	p	IRR
Intercept	2.835	1.555	0.068	17.03
log(area)	−1.551	0.239	8.5e-11	**0.212**

LR test, offset-only → offset+slope: $\chi^2 = 32.05$ on 1 df, $p = 1.5 \times 10^{-8}$.

H3 therefore survives. A doubling of LGA area multiplies density by $2^{-1.551} = 0.342$ — i.e. density falls by about 66% per doubling of area. The direction, the significance, and the existence of the effect are all confirmed under a specification that cannot produce negative densities and does not divide by the denominator.

Honest caveat on the hurdle model

The hurdle has the lowest AIC but **its count component is not identified**. Only 39 of 775 LGAs are non-zero, and a zero-truncated negative binomial must estimate θ from those 39 alone; θ collapses to the boundary and the standard errors return `NaN`. Its AIC is therefore not comparable in the usual way, and its count coefficients must not be quoted. Only the **zero component** is stable: $\log(\text{area})$ coefficient -0.813 ($p = 2 \times 10^{-10}$), meaning larger LGAs are more likely to sit outside the mapped set entirely — which is a statement about **mapping coverage**, not about agriculture.

6.3.c 6.3.3 LISA with multiple-comparison control

The earlier analysis flagged every LGA with raw $p < 0.05$ as a cluster: **23 of 775**. But with 775 simultaneous tests the expected false-positive count is **38.8**, so reporting 23 without adjustment was not defensible. Observing *fewer* than chance expectation is itself the signature that the normal approximation behind the LISA p-values is **conservative** on a variable that is 95% zeros — the p-values are discrete and understated, not the clustering absent.

Criterion	LGAs significant
Raw $p < 0.05$ (uncorrected)	23
Benjamini–Hochberg FDR 5%	**19**
Bonferroni	18
Expected false positives at $\alpha = 0.05$	38.8

The reportable figure is the **BH-adjusted 19**. A join-count test on the binary "has any mapped feature" indicator (`07_lisa_fdr_pointpattern.R`) gives $z = 30.3$, $p < 2 \times 10^{-16}$, and within-city Ripley's L lies outside its simulation envelope in all five cities.

But note what the adjusted LISA set actually contains: 18 of the 19 significant LGAs also have $n > 0$, and 18 of the 39 LGAs with any points are significant. The LISA result and "this LGA is in the mapped set" largely coincide — which is the OSM-completeness confound reappearing inside the spatial test, and the reason the exposure control in §6.5 matters.

6.3.d 6.4 The climate hypothesis (H2) cannot be tested with these data

The earlier text reported PRECTOT significant at $p = 0.018$ (spatial error) and concluded H2 was supported. That conclusion is withdrawn. The audit's arithmetic:

- The 34 LGAs carry **5 distinct rainfall values** — one per city.
- Regressing rainfall on city dummies gives $\boldsymbol{R^2 = 1.000}$: rainfall *is* city identity.
- The subset's spatial weights form **5 connected components with one isolate**, so the "spatial" term absorbs city effects rather than modelling inter-LGA spillover. The fitted $\rho = -0.438$ and $\lambda = -0.476$ are negative, which is meaningless here.
- Between-city variance is only **ICC = 0.102** — 90% of the outcome variation is *within* cities, while the covariate varies only *between* them.

Refitting at the level the data actually replicate:

Specification	rainfall b	p
Naive OLS, 34 rows	−0.000085	0.250
Spatial lag, 34 rows	−0.000135	0.035

Specification	rainfall b	p
Spatial error, 34 rows	−0.000100	**0.018** ← previously reported
Collapsed to 5 city means	**-0.0000438**	**0.576**

Spearman at the city level: $\rho = -0.20$, $p = 0.78$.

Conclusion: H2 is untestable with the frozen climate data. The honest statement is that the data are *consistent with* the aridity hypothesis — Kano (404 mm) has the highest mean density (0.204 km^{-2}) and Port Harcourt (1,891 mm) among the lowest (0.069) — but with $K = 5$ and 3 residual degrees of freedom no effect can be established, and the two-city contrast is not evidence.

To test H2 properly requires rainfall with *within*-city variation, which rasterises to LGA polygons: **CHIRPS** (~5 km), **ERA5-Land** (~9 km) or **TAMSAT** (4 km). Any of these gives each LGA its own value, turning 5 distinct predictor values into 34. Until then this remains an open question, not a finding.

> In plain English
>
> The mistake was subtle and common: rainfall was measured once per city, then handed to every LGA in that city. The model then behaved as if it had 34 independent observations of rainfall when it really had 5. More rows do not create more information. When the analysis is run at the level the data actually support — five cities — the rainfall effect vanishes.
>
> *What survives:* the area effect (H3), now confirmed under a count model, and the spatial dependence (H1), which is strong and robust.

6.3.e 6.5 The OSM-completeness confound (open)

`landuse=farmland` inside a city bounding box measures how much of a place OpenStreetMap contributors have mapped, at least as much as it measures urban agriculture. The two are entangled: Kano is the best-mapped state in Nigerian OSM *and* has the most "urban agriculture" points (189 of 484).

`scripts/fetch_osm_exposure.py` measures the confound directly by querying per-LGA `way["building"]` and `way["highway"]` counts inside each LGA polygon via Overpass `out count;` (aggregate counts, so responses are tiny and the run is resumable). Treating the building count as an **exposure offset** asks: given how much of this LGA is mapped at all, is urban agriculture unusually present or absent?

Results land in `results/tables/`. This is **not resolved**; it is measured and reported.

7 Additional analyses

7.1 7.1 Density vs climate (descriptive only) and scale

Revised 15 Oct 2026 — the correlation figures below inherit the same defect as §6.4 and are now descriptive only. The earlier text read temperature $\rho = +0.44$ ($p = 0.009$) as support for H2. That is not supportable: the 39 (mapped: 34) LGAs carry only **5 distinct rainfall and temperature values**, because both were sampled at city centroids and assigned to every LGA in that city. Spearman on such a variable measures the ordering of five cities, not of 34 LGAs, and any p-value from it is meaningless.

At the honest city level ($K = 5$): rainfall $\rho = -0.20$, $p = 0.78$.

So these figures are retained as **descriptive city contrasts** and labelled as such: `density_vs_rainfall.png` (scatter with the five city values visible) and `climate_correlation.png` (matrix). Their captions state $K = 5$ and that no inference is drawn.

What *does* survive in this section is the **scale** comparison, which uses LGA-level quantities and is unaffected: Tarauni 23.6 km² at 1.06 km^{-2} versus Ungogo 187 km² at 0.48 km^{-2} supports H3, and the top-10 table shows why counts mislead — Ikorodu has 53 points but only 0.14 km^{-2} over 369 km². The corrected H3 model is in §6.3.2.

7.2 7.2 Maps

`analysis/spatial/03_maps.R` + `04_extra_figs.R` produce `results/figures/osm_points.png` (484 points by city), `lisa_map.png` (HH red, LL blue), `density_hist.png` (zero-inflated, median 0, mean 0.62), `density_vs_rainfall.png`, `climate_correlation.png`, `density_by_state.png` (Kano median 0.15 km^{-2} vs Lagos 0.14, Gombe 0.016, FCT 0.005) — all `ggplot2`/`sf`, 300 dpi.

8 Reproducibility, ethics, reflection

Reproducibility: Every number traces to `data/raw/provenance.json` (timestamped URL/licence) and version-pinned code; `git log` shows `scripts/fetch_*.py` → `data/raw` → `R` → `quarto` → `manuscript.docx/docs/methodology.pdf`. No network in `R` — only frozen CSVs.

Ethics and positionality: Open data (ODbL OSM, CC BY GADM/World Bank, NASA POWER) with attribution; no PII; public allotment centroids; auditable via `provenance.json`; authors have no residential positionality in LGAs, so inference requires community validation before zoning.

Process reflection (5 phases): Phase 0 installed missing `spatialreg`; Phase 1 staged Overpass 406⟶504 per-tag fixes to obtain 484 points (isolated Gombe 45 test → full 189/130/88/45/32); Phase 2 found 775 vs nominal 774 LGAs (GADM version artefact) and deleted `NGA_0/1` to keep repo 4.7 MB; Phase 3 corrected `NAME_2`⟶`GID_2` bug (775⟶39 LGAs, 5%) and showed I 0.158⟶0.300 and queen–rook 0.300⟶0.306 robustness; Phase 4–5 embedded 3–6 figs and trimmed 4,459⟶2,307 w for the earlier 12-page conference limit (the AJLP&GS manuscript is longer). Iterative z 8.47⟶17.61 demonstrated area normalisation is decisive.

9 How to re-run

```
# Python harvest (one-time)
conda activate C:\Users\TOSHIBA\ds-general
C:\Users\TOSHIBA\ds-general\python.exe scripts/fetch_worldbank.py --out data/raw/
worldbank_ng.csv
C:\Users\TOSHIBA\ds-general\python.exe scripts/fetch_osm_urban_ag.py --city Gombe
--out data/raw/osm_gombe_test.csv
# ... 5 cities ...

# Download GADM LGA (if not present)
# https://geodata.ucdavis.edu/gadm/gadm4.1/shp/gadm41_NGA_shp.zip → data/external/
gadm41_NGA_2.*

# R pipeline
Rscript analysis/spatial/01_clean.R          # → data/processed/*.gpkg
Rscript analysis/spatial/02_spatial_stats.R  # → results/lga_lisa.gpkg (Moran/LISA)
Rscript analysis/spatial/03_maps.R
Rscript analysis/spatial/04_extra_figs.R
```

```
Rscript analysis/spatial/run_regression.R
Rscript analysis/spatial/check_sensitivity.R

# Render
quarto render manuscript/manuscript.qmd --to docx  # the manuscript (AJLP&GS format)
python scripts/fix_docx_ajlpgs.py                  # Cambria 12pt / 1.2 / AJLP&GS
margins + author block
python scripts/verify_author_block.py              # fail if any output lost the
author block
quarto render docs/methodology.qmd --to typst        # this file
quarto render manuscript/manuscript.qmd --to html --to pdf
```

Page count check:

```
# docx words → pages estimate
python     -c     "import     zipfile,re,pathlib;     d=pathlib.Path('manuscript/
manuscript.docx');         w=len('         '.join(re.findall(r'<w:t[^>]*>([^<]+)</
w:t>',  zipfile.ZipFile(d).read('word/document.xml').decode())).split());  print(w,
round(w/250,1), round(w/300,1))"
# Word COM actual pages
python -c "import win32com.client; w=win32com.client.Dispatch('Word.Application');
d=w.Documents.Open(r'D:\urban-agriculture-
nigeria-spatial\manuscript\manuscript.docx');       print(d.ComputeStatistics(2));
d.Close(False); w.Quit()"
# PDF actual pages
soffice --convert-to pdf manuscript/manuscript.docx; python -c "from PyPDF2 import
PdfReader; print(len(PdfReader('manuscript/manuscript.pdf').pages))"
```

10 References

Bibliography

[1] Food and Agriculture Organization of the United Nations and International Fund for Agricultural Development and United Nations Children's Fund and World Food Programme and World Health Organization, *The State of Food Security and Nutrition in the World 2023*. FAO; IFAD; UNICEF; WFP; WHO;, 2023. doi: 10.4060/cc3017en.